Inverting glacier history for past temperature

A differentiable glacier model applied to the Little Ice Age at the Great Aletsch Glacier

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Past climate is heavily underconstrained

Temperature proxies are sparse, and most of them speak for a global or a hemisphere — not for regions where a glacier sits.



Ice cores

Greenland, Antarctica

Footprint
Hemispheric, high-latitude

Time span
Millennia



Speleothems

and lake sediments

Footprint
Regional; mixes temperature and moisture

Time span
Millennia



Instruments

Station network

Footprint
Local

Time span
Only since the mid-1800s



Glaciers

Mapped extents, moraines

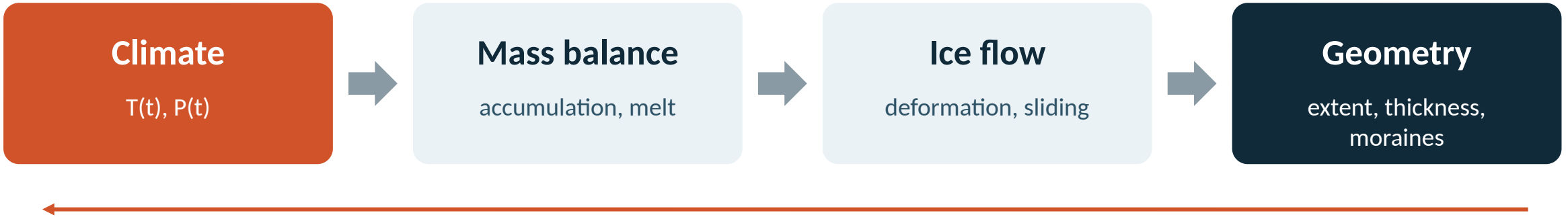
Footprint
One valley

Time span
Centuries (moraines: millennia)

Glaciers are local and long-lived, and well documented — **but they record climate through ice dynamics, so the record has to be inverted.**

A glacier is a slow, indirect thermometer

forward model: well understood



inverse problem: what we want

Lag

The tongue responds to climate over 100–200 years, so today's extent reflects past climate.

Distant forcing

Ensembles drive their parameters with distant records (North Atlantic, Greenland) that disagree with local observations.

Non-unique

Warmer melts like drier: temperature and precipitation trade off against each other.

Ensembles: cost grows with every parameter

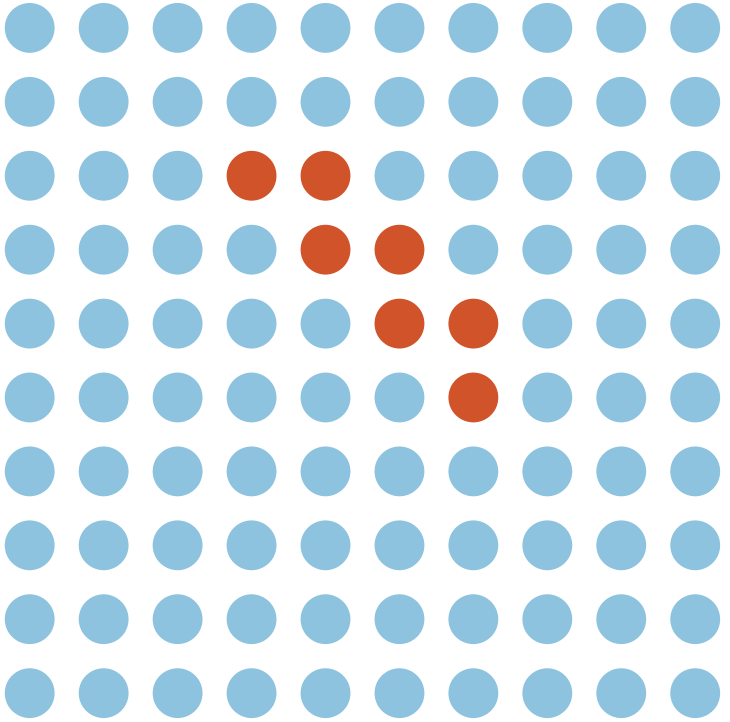
The usual inverse: sample parameters, run the forward model many times, keep the runs that fit.

One-at-a-time sensitivity
≈ 2 forward runs per parameter, per perturbation size

Grids & Monte Carlo
cost multiplies with every added dimension

So climate gets compressed
into a few numbers: an offset, a step, a trend

And data get thinned
one observation type at a time, hand-tuned

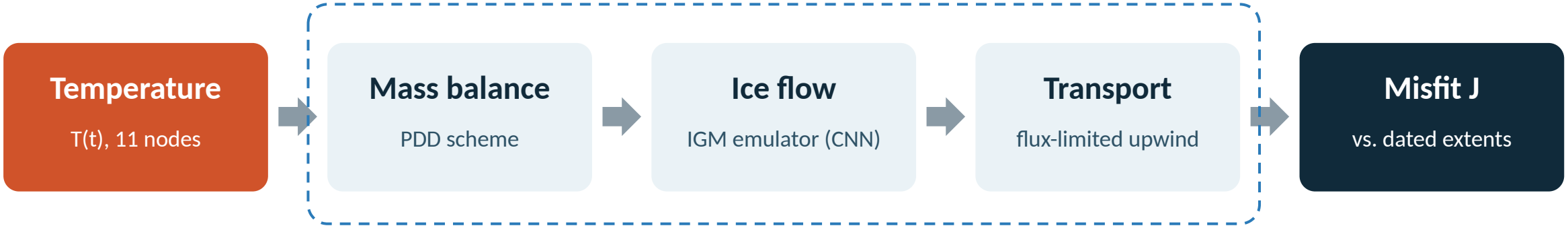


100 forward runs
7 accepted — and that is only 2 parameters

11 temperature nodes × 5 values = **$5^{11} \approx 49$ million runs**

Our approach: a differentiable glacier model

repeated every time step · ~5,000 steps from 1500 to 2017



Backward pass: automatic differentiation carries $\partial J/\partial T(t)$ back through every time step

Real ice physics

PyTorch port of IGM's pretrained ice-flow emulator; matches the TensorFlow original to 2.4×10^{-6}.

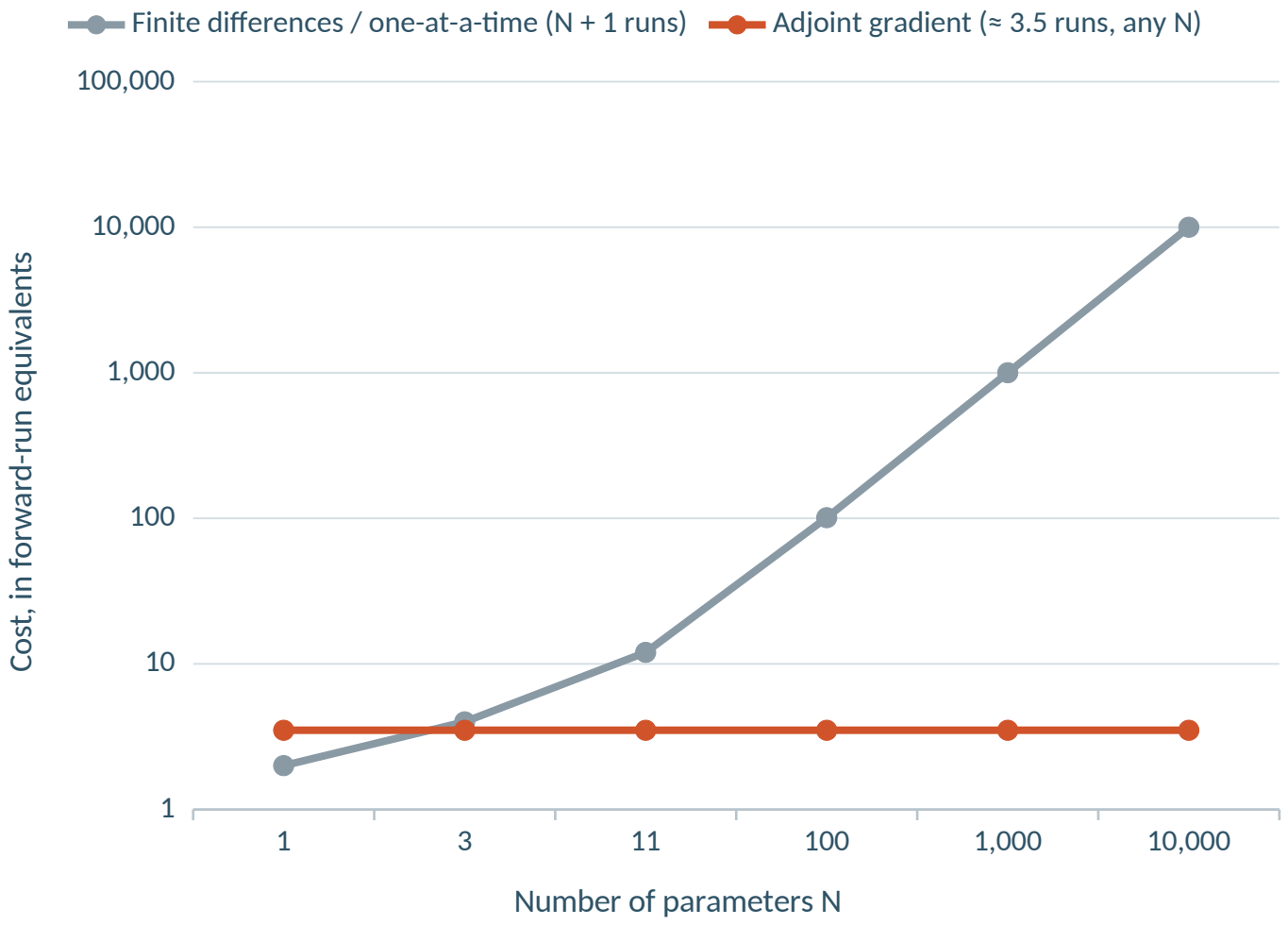
Fits in memory

Gradient checkpointing recomputes each step on the way back, so 517 years run on one GPU.

Gradients verified

With a fixed time step, the adjoint matches finite differences exactly.

One gradient gives every sensitivity



≈ 3.5

forward runs for one gradient: 173 s per forward + backward vs. ~50 s per forward (1500–2017, RTX 4090)

~3 h

60 iterations invert the full 11-node temperature history

≈ 9,000

ice-covered cells: a spatial field (e.g. sliding τ_ref) costs one gradient too, vs. ~5 GPU-days of runs

Which observations can enter the loss?

$$J(T) = \sum_k w_k \cdot \text{misfit}_k(\text{model}, \text{observation}_k) + \lambda \cdot \text{smoothness}(T)$$

Dated extents

Historical maps and DEMs, compared through a soft (sigmoid) ice mask so the margin stays differentiable.

In use: 5 epochs

Surface elevation & moraine crests

Same DEMs (volume −34 % vs. area −6 %), and moraine crest heights, already built for the Younger Dryas.

Ready

Luminescence data → temperature?

Could luminescence thermometry of exposed rock constrain past temperature?

To discuss

Erosion rates → precipitation?

Could catchment erosion rates constrain precipitation or ice flux? Exposure ages could date moraine stands.

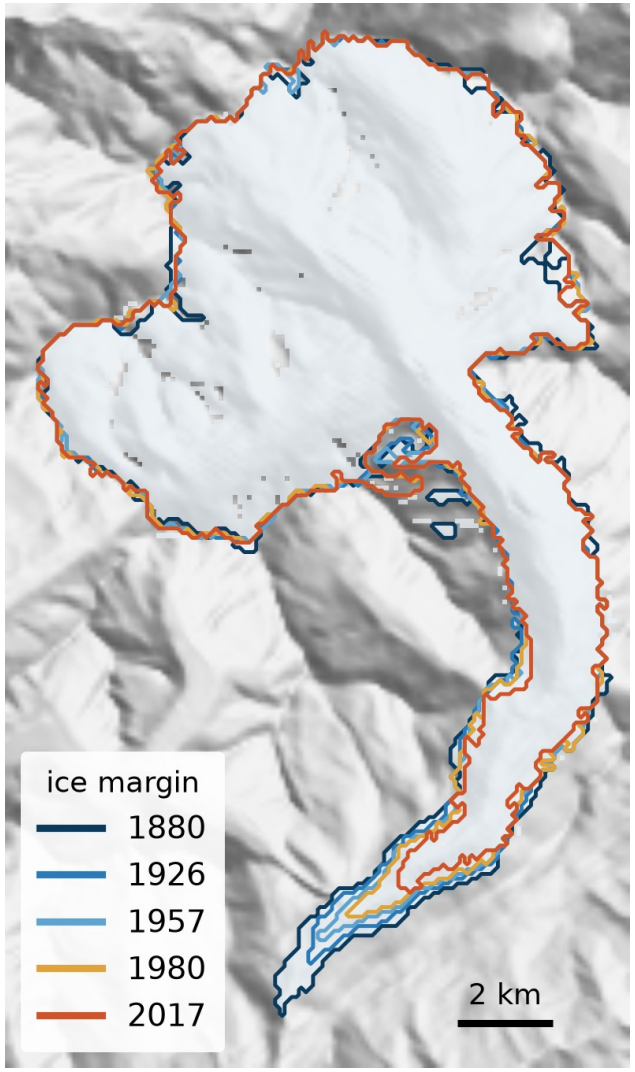
To discuss

APPLICATION

Little Ice Age temperature at the Great Aletsch Glacier, 1500–2017



Aletsch: 140 years of dated surfaces



7

dated surface DEMs, 1880–2017 (5 used: 1880, 1926, 1957, 1980, 2017)

100 m

model grid, 225 × 132 cells; started from the 2017 geometry

−34 %

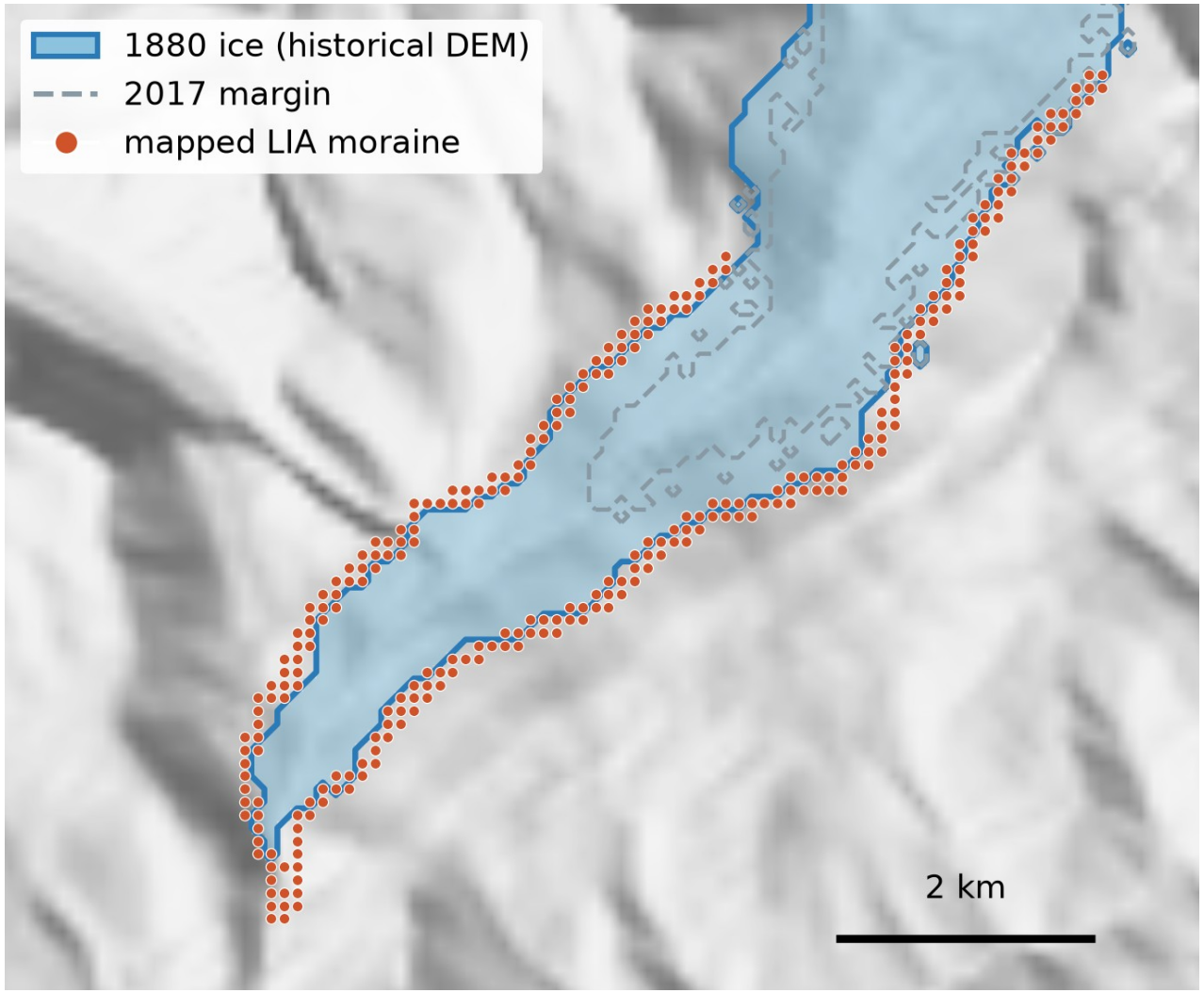
ice volume, 1880 → 2017 (20.7 → 13.7 km³)

−6 %

ice area over the same period (96 → 91 km²)

Aletsch thins far more than it shrinks. Extent is the information we are allowed to use, terminus-weighted signal — the inversion has to work from the tongue.

Independent check: LIA moraine = 1880 margin



100 m

median distance from the mapped LIA moraine to the 1880 ice margin — one grid cell

86 %

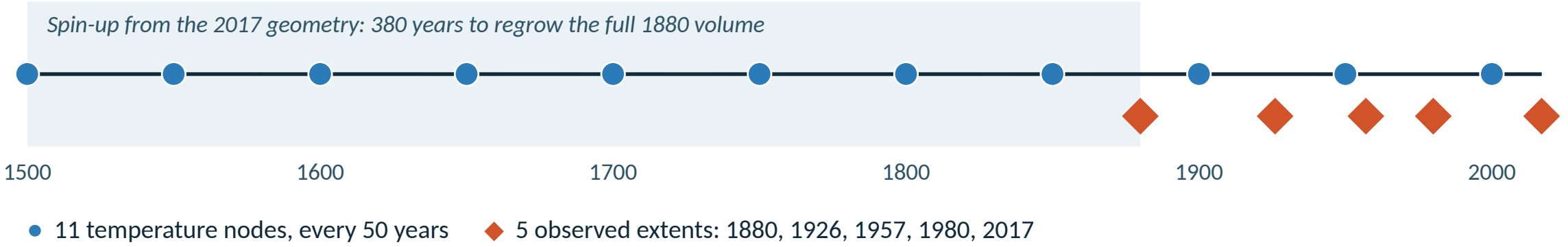
of moraine cells within 150 m of the 1880 margin

505 m

median distance to the 2017 margin — the offset grows with every later epoch

Two independent datasets agree: this validates the 1880 DEM and calibrates the moraine operator we will need for older, undated stands.

Experiment: start from today, run 517 years



Control

T_lowest(t): mean annual temperature at the lowest grid point (799 m), lapse rate 7 °C/km

Objective

Soft-extent misfit, averaged over the 5 epochs, plus a smoothness penalty on T(t)

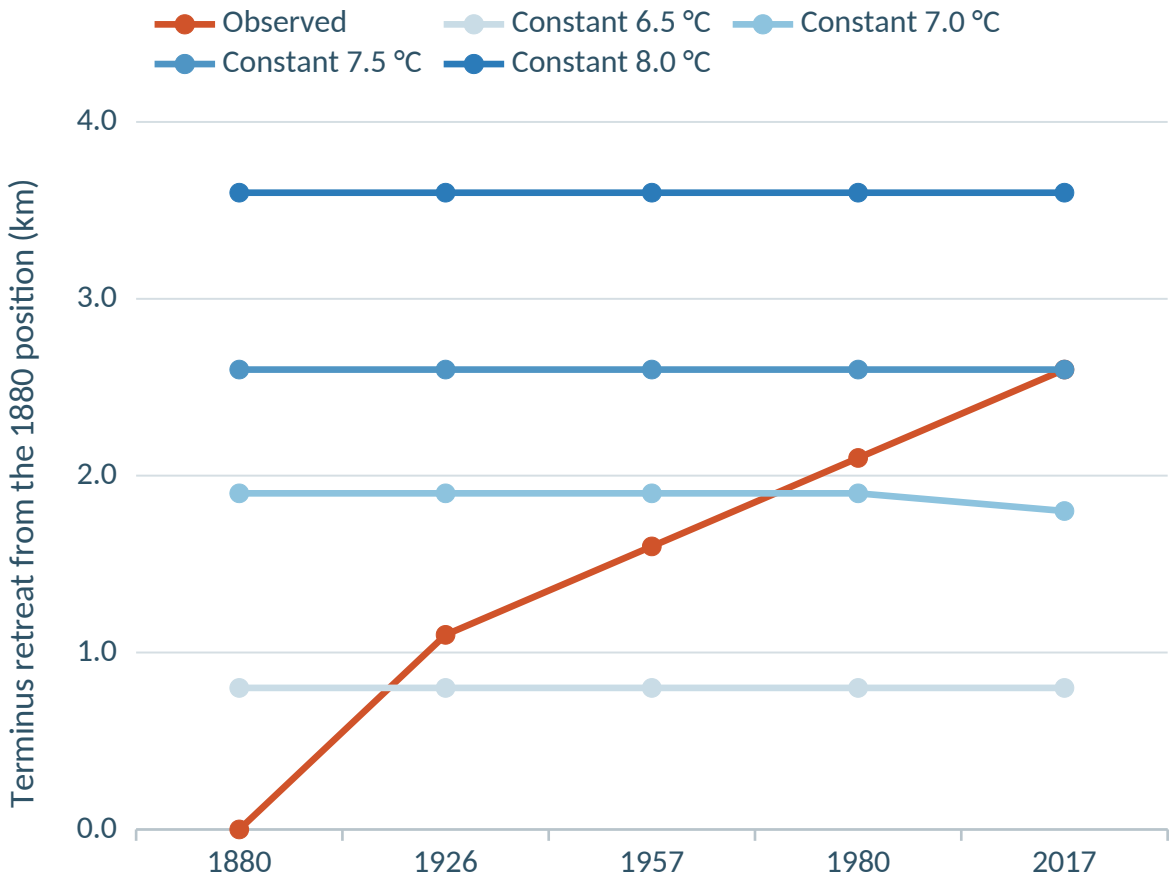
Held fixed

Melt factor 0.225, precipitation 1.8 m/yr, τ_ref 0.215 MPa,
$A = 78\ \mathrm{MPa}^{-3}\ \mathrm{yr}^{-1}$

Optimiser

Adam, learning rate 0.02, 60 iterations, one RTX 4090

Constant climate cannot explain the retreat



Constant climate: the modelled tongue settles and stays put. Observed: 2.6 km of retreat.

Where the +1.36 °C comes from

1 Four runs at constant 6.5, 7.0, 7.5 and 8.0 °C. Each glacier settles at one equilibrium terminus (the flat lines).

2 Place each observed terminus on that curve and read off the temperature that would hold it there in equilibrium.

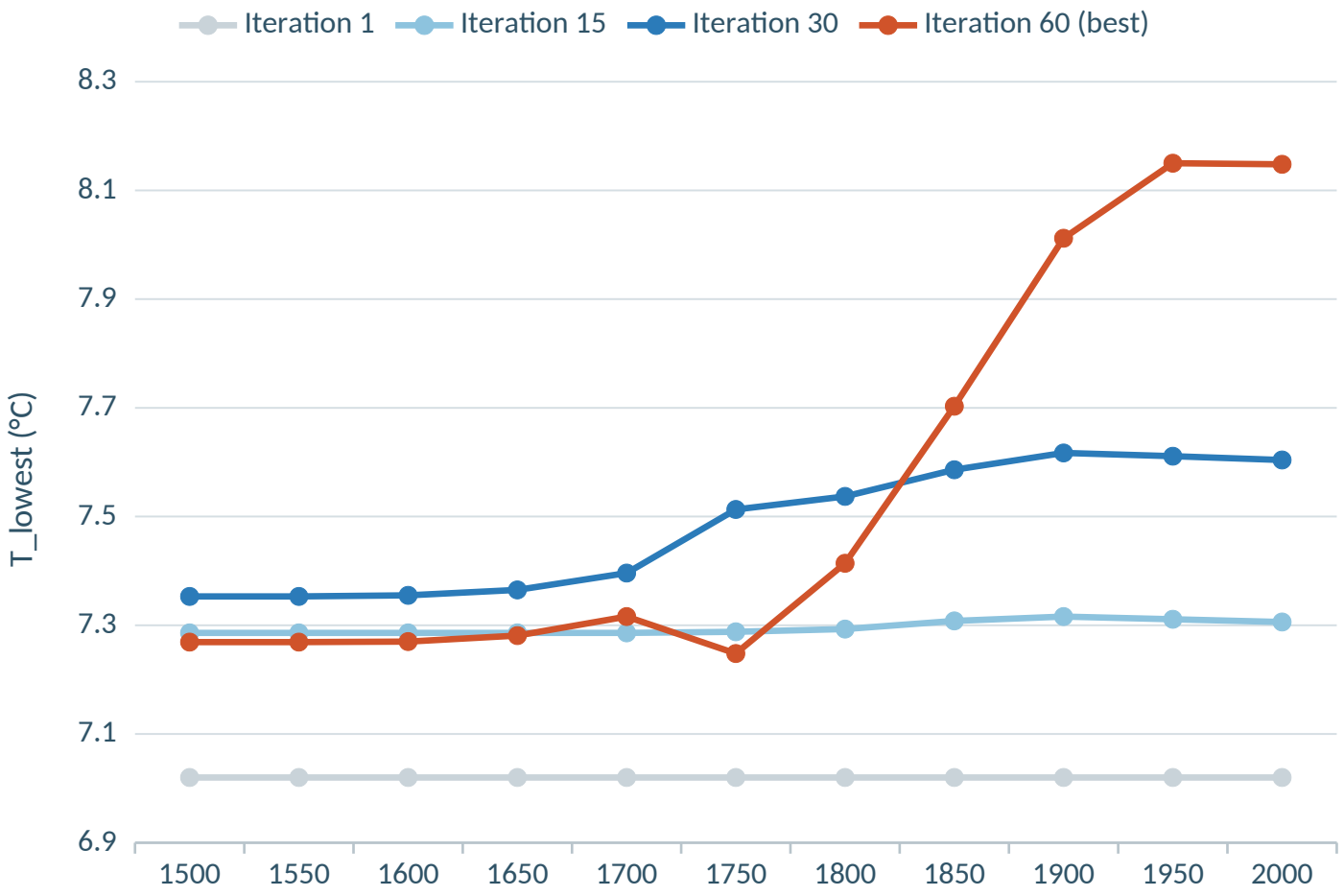
1880	1926	1957	1980	2017
6.14*	6.64	6.86	7.14	7.50

3 Difference, 2017 − 1880 (7.50 − 6.14):

+1.36 °C warming implied by the termini

** Extrapolated: the 1880 terminus lies 0.8 km beyond the coldest run. Each epoch is treated as if in equilibrium; the 2017 glacier lags its climate most, so the true warming is probably larger. A lower bound by argument, not by measurement.*

The inversion finds +0.9 °C of warming



1500–1650: no observation within a response time — shaped by the smoothness prior only.

+0.90 °C

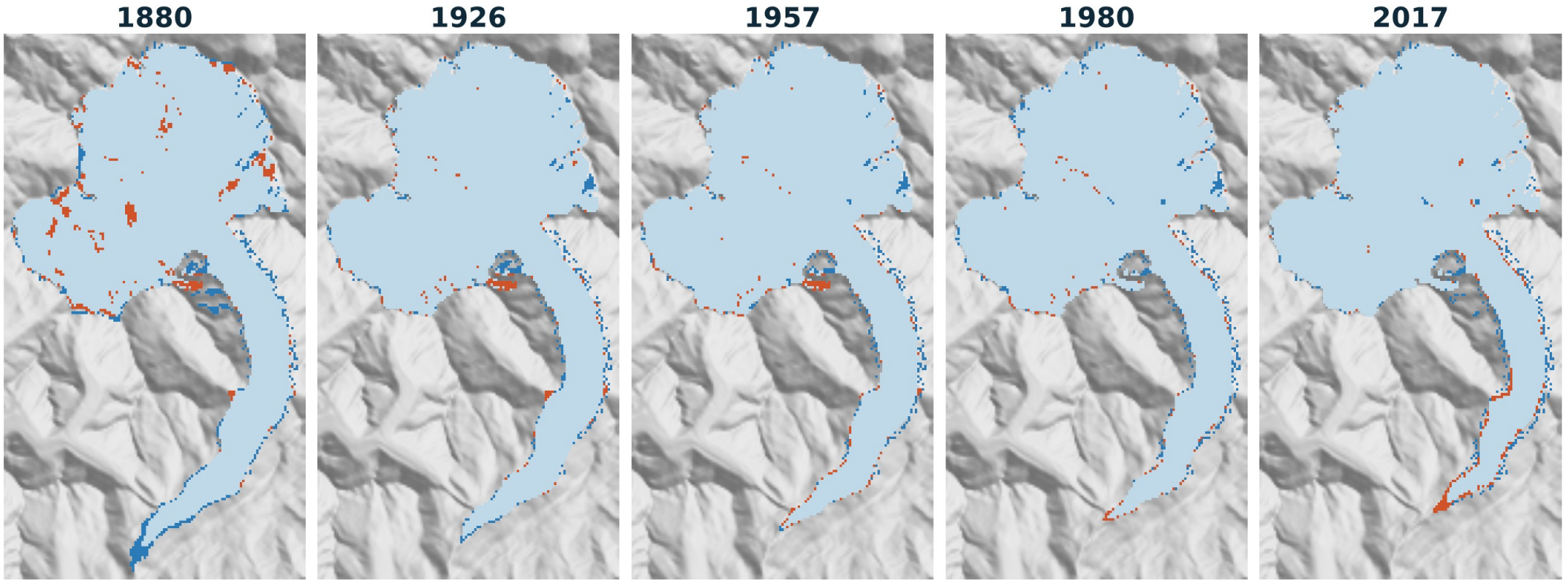
from 1750 to 1950, recovered from extent alone

0.90 → 0.94

mean extent overlap (IoU) across the 5 epochs; loss down 47 %

T_lowest is a model parameter, not a station temperature: the trend is the interpretable quantity.

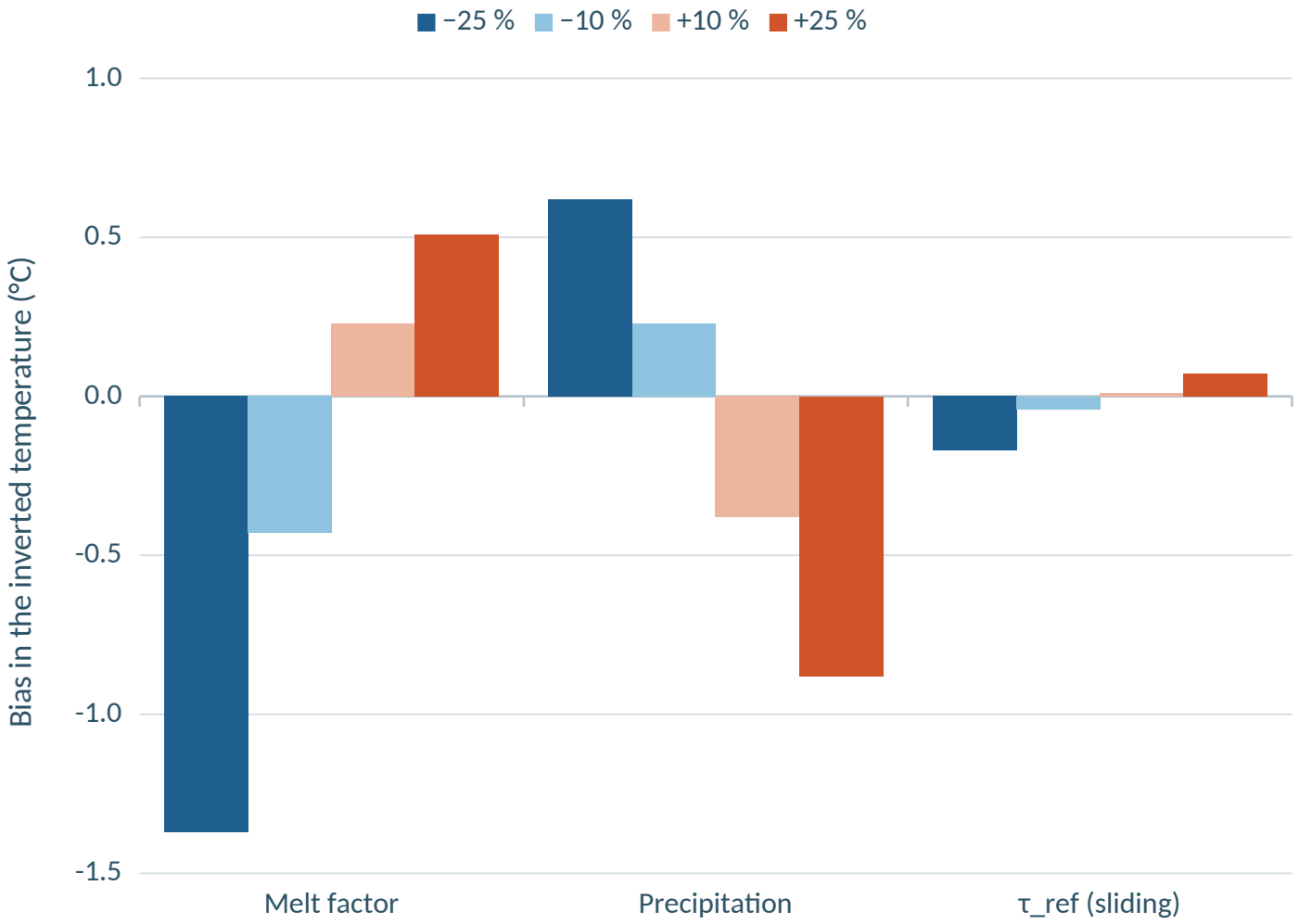
Simulated extents match all five epochs



Residuals: a one-cell band at the terminus, and in 1880 thin ice on steep headwalls that the real glacier sheds by avalanching.

Open question: mask the steep headwalls (tested: removes 96 % of this misfit), and/or add pre-1880 terminus positions from historical reconstructions?

Mass balance, not ice flow, sets the answer



If a fixed parameter is off by this much, the inversion returns a temperature this far off to compensate.

0.2–0.4 °C

bias from a 10 % error in melt factor or precipitation

≤ 0.05 °C

from the same error in sliding (τ_ref): 5–15× smaller

Error bars on T(t) are set by the mass-balance priors. Melt and precipitation trade off, so extent constrains roughly their ratio.

What we learned about the inverse problem

~20 iterations

Mean before trend

All 11 node gradients initially share one sign, so the optimiser fits the mean temperature before any trend emerges. Parameterising level and shape separately removes this delay.

36 %

Structural model error

of the extent misfit (60 % in 1880) lies on ~34° headwalls above every attainable ELA, where the real glacier loses ice by avalanching. No climate can remove it, yet it steers the gradient.

< 0.01

Extent is a weak signal

change in mean IoU per °C. Aletsch lost 34 % of its volume but only 6 % of its area since 1880, so many climate histories fit the outlines almost equally well.

Implication: a credible T(t) needs a converged trend, an explicit treatment of structural error, and observations beyond extent.

Conclusions and next steps

What we showed

1. A differentiable glacier model inverts a 517-year temperature history from 5 dated extents in ~3 GPU-hours.
2. Aletsch needs warming: +0.9 °C between 1750 and 1950 recovered; ≥ +1.36 °C implied by the termini.
3. One gradient costs ~3.5 forward runs for any number of parameters: sensitivity without ensembles.
4. Mass balance matters 5–15× more than ice flow for the answer.

Next



Add surface elevation

volume −34 % vs. area −6 %: far more signal than extent



Invert temperature and precipitation jointly

test how much of the trend is accumulation in disguise



Error bars

gradient-informed ensembles over the melt and precipitation priors



Younger Dryas

421 dated moraine cells (11.8–14.2 ka) already on the model grid

Thank you — questions?

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